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Acknowledgement

We would like to express our deepest gratitude towards Islington College and The London Metropolitan University who granted us the academic basis, resources and collaborative space to make this project feasible. With practical skills in Cloud Computing and IoT developed within the curriculum and facilities, the technical implementation of this work would have been impossible without this contribution.

We also owe much to our module leader Mr. Aayush Bhakta Pradhanang, who so skilfully guided us and organized a module that taught us the much needed theoretical and practical knowledge in taking up this venture. His experience with system architecture and information of IoT was priceless. We are also grateful to our tutor, Mr. Bishnu Pandey, who has remained supportive, provided with constructive feedback at several stages of development, and continually encouraged us to overcome obstacles and optimize the results of our project, which served as significant assistance in overcoming all difficulties and improving the results of our work.

Lastly, we recognize the combined effort of our team members whose diligent work, working team and technical input played a critical role in the completion of the project. This report is the result of the acquired experience of our institution and mentors, as well as our learning experience with our team and our diligent work.

Abstract

A safety device is an IoT-based device that is designed for enhancing women's personal safety in emergency situations. This device supports the generation of warnings in a quick manner with the help of automated SMS to pre-programmed contacts based on the ESP32 microcontroller. The notifications also provide the present position of the user which is developed by the embedded GPS and GSM modules.

The device sends an alert signal in sync to scare away predators and draws attention of people passing by to seek assistance. The system is portable and easy to use, which makes it possible to get assistance anywhere, any minute, and do it in an efficient and fast manner

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1. Introduction

1.1. Current Scenario

The safety and security of women is one of the most pressing issues that are yet to be addressed. The figures are shocking across the world. A report by the World Health Organization (2021) estimates that 31 percent of all women throughout the world, or about 736 million individuals, have faced physical and/or intimate partner violence or non-partner sexual violence in their lifetime, although there has been a slight rise in this figure in the last ten years. This does not just represent a statistic but a deep sign of vulnerability in a system.

The situation in Nepal is especially severe and requires certain consideration. The Nepal Health Research Council (2022) survey Violence against Children and Youth (VACS) found that 17 percent of females and girls aged 13-24 years had experienced sexual violence in childhood and 31 percent of all women who ever married aged 15-49 years experienced physical violence. These national statistics highlight a grim situation in which the old protective regimes are in failure and the interventions of technology are not appropriately adjusted to local surroundings and circumstances of extreme crisis.

Current technological solutions mainly smartphone applications are limited by tremendous functional gaps during an emergency. They demand that the user should unlock a phone, use an application, and have a reliable network connection- things that are not usually feasible in the middle of an attack. Moreover, a smartphone is conspicuous, can be readily snatched, or rendered useless, and one may run out of battery. The dependency on a multitasking gadget generates a risky latency and reliability margin, because of which women do not have a specific, immediate, and safe way to request assistance.

1.2. Problem Statement and Project as a Solution

Problem Statement: The lack of a dedicated, fast deployable and reliable hardware-based emergency alert system in women that is not dependent on a smartphone is in critical need. Present options are not as immediate, discrete and assured to work at the specific time of danger posed, which poses an extremely dangerous lag in help.

Project as a Solution: This technological and safety gap is directly addressed in this project by designing a specific IoT-based safety equipment proof-of-concept. The answer is an in-built electronic mechanism that is quick and should be activated with a single touch. It uses the Internet of Things (IoT) paradigm to design a self-sufficient unit, which, when triggered by a manual discrete step or by an automated fall detection system with the help of an inertial sensor, triggers a deterministic multi-layered alert protocol.

The main innovation is its freedom to operate; the device will automatically obtain its geographical position through an in-built Global Positioning System (GPS) device and send an immediate structured Short Message Service (SMS) notification with a live Google Maps link to programmed emergency contacts through a built-in Global System for Mobile Communications (GSM) unit. At the same time, it will also trigger a local piezoelectric audio alarm that will work as a direct deterrent and draw attention to the bystanders. The project will be used to test the viability of a sound hardware-software architecture in personal safety by creating, coding, and testing this working prototype.

It goes beyond mobile applications to prove a physical, quicker and more dependable way of women signalling stress, therefore providing a guideline upon which a subsequent commercially viable and a life-saving wearable device can be established.

1.3. Aims and Objectives

Aim: The main aim of the project is to design, develop and empirically test a functional proof-of-concept of an IoT-based personal safety device mechanism , and create a stable technical platform of immediate emergency alerts and real time location broadcasting.

Objectives:

- To design and develop a strong piece of hardware that incorporates an ESP32 microcontroller, a NEO-6M GPS receiver, a SIM800L GSM modem, an MPU-6050 inertial measurement unit, an alert interface.
- To create, debug, and install embedded firmware that enables dual triggering of emergencies (through a manual button press and an algorithmic fall detection) and control the resulting automatic response of an alert.
- To introduce and test a robust communication protocol that would guarantee a successful delivery of formatted SMS alerts that would have the right GPS location to various identified contacts.
- To test extensively the functionality and performance of the integrated prototype, to quantitatively test the prototype in terms of alert latencies, GPS accuracy, the SMS delivery success rate and general stability of the system at different conditions.

2. Background

2.1. System Overview

The Safety device is a high tech IoT based device that is designed to offer a rapid, independent emergency signalling to individual security. The device is a self-sufficient device designed to substitute smartphone-based safety applications and can that detect emergencies via two triggers: A covert tactile push button and an automatic fall detector - and automatically notifies predefined emergency contacts with an SMS containing real-time GPS coordinates. (U-blox, 2021)A local piezo buzzer provides an audible warning and attracts immediate attention. The system helps in case of life-threatening circumstances where the mobile phone will be inaccessible or dead.

Key Features:

- **Dual-Trigger Activation:** The device can be used in both manual and automatic mode, which is done through a tactile push button used as a covert communication and an accelerometer applied to detect a fall.
- **Autonomous Emergency Notification:** Once activated, the system automatically delivers emergency contacts (predefined) SMSs without the need of using a second smartphone connection via the GSM module (SIM800L / SIM7000).
- **Real-Time Geolocation:** An inbuilt GPS system (NEO-6M) is also able to give the exact coordinates of the user in the emergency SMS, which would then be used to find the individual with a high degree of precision by the responders.
- **Localized Deterrent and Alert:** A high frequency piezo buzzer is a local alarm that attempts to prevent the possible predators and give attention to nearby passersby in the event of a crisis.

Core Technologies:

- **Central Processing Unit:** ESP32 is the brain of the system, which reads sensor data and performs the logic of the firmware that is needed to execute the emergency response.
- **GPS and GSM Connections:** The NEO-6M GPS module will be able to receive real-time location information, and the SIM800L or SIM7000 GSM module will enable the cellular connection to send out SMS alerts.
- **Motion Sensing:** The MPU-6050 accelerator and gyro sensor will offer them the motion-sensing options they require to detect falls and unexpected hits.
- **Power:** Li-Po 3.7 V battery with USB charge circuit, battery tracking and low battery notifications.

2.2. Design Diagrams

2.2.1. System Architecture

System architecture in IoT The system architecture of an IoT system is the general structural design of an IoT system, the manner in which various components of the system (e.g. the devices, the networks, data processing units and applications) are structured and co-exist. (Sharma, 2022) It describes the origin of data, their transmission, processing, data storage and the ultimate display to the users with scalability, security, and effective communication. The architecture of the IoT systems tends to be stratified and assists developers with realizing the entire process of the system, both in physical sensing and decision making at the user level.

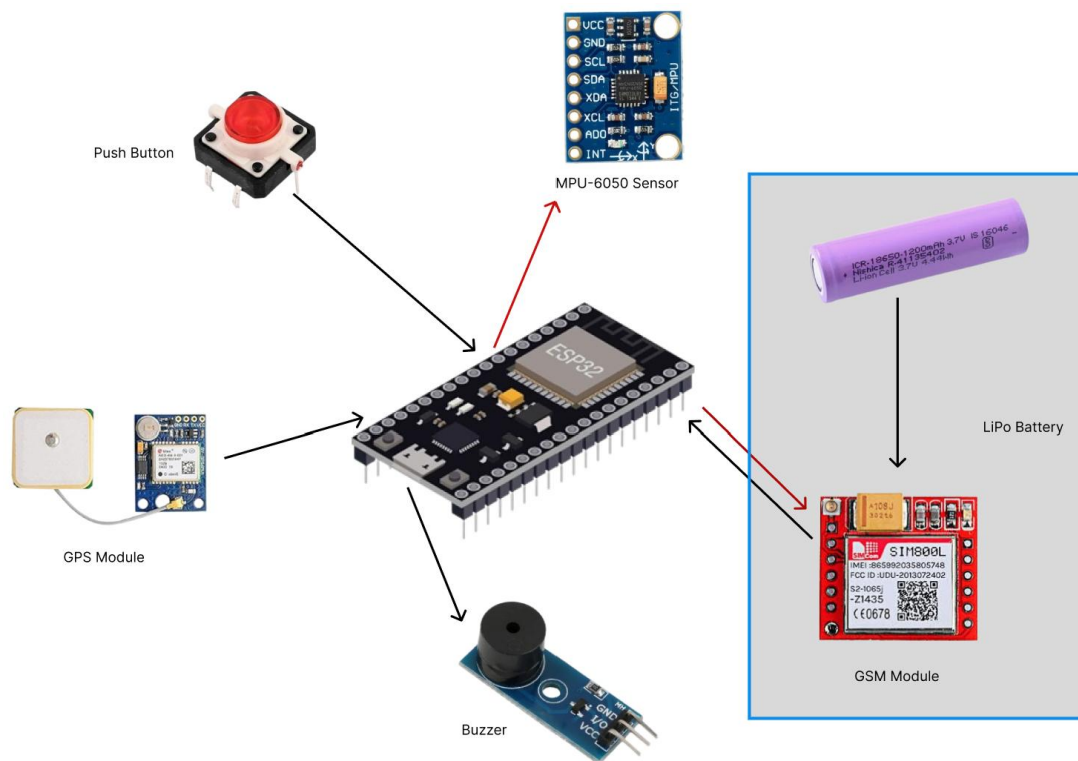


Figure 1: Pictorial Hardware Architecture

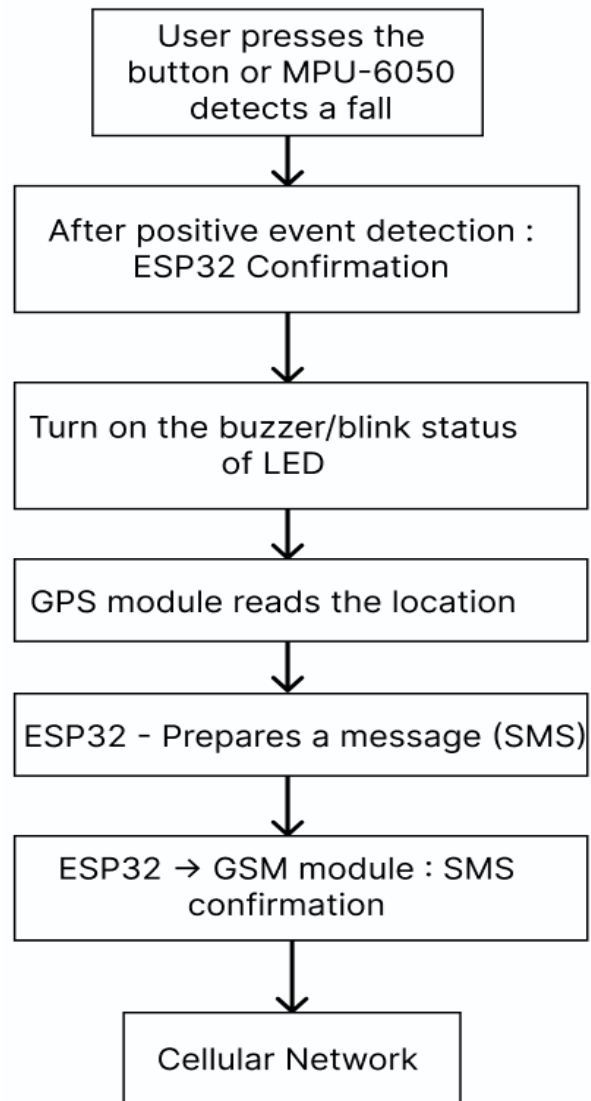


Figure 2: Detailed System Architecture

2.2.2. Flowchart

A flow diagram is a graphical composition of a process, system, or algorithm and it is one of the necessary tools to draw the complex logic in a graphical form. It also makes it easy to know the architecture of the system in that a step-by-step solution to a problem or a procedure is provided. (McRoberts, 2020) Flowcharts are especially useful in the field of engineering and software development when it comes to large-scale projects because they allow the team to see how an internal work of a system runs and in what sequence. Standardized symbols are used to represent certain actions: ovals are used to depict the starting and finishing point, rectangles are used to depict processing steps, diamonds are used to depict decision points and parallelograms are used to depict input or output operations.

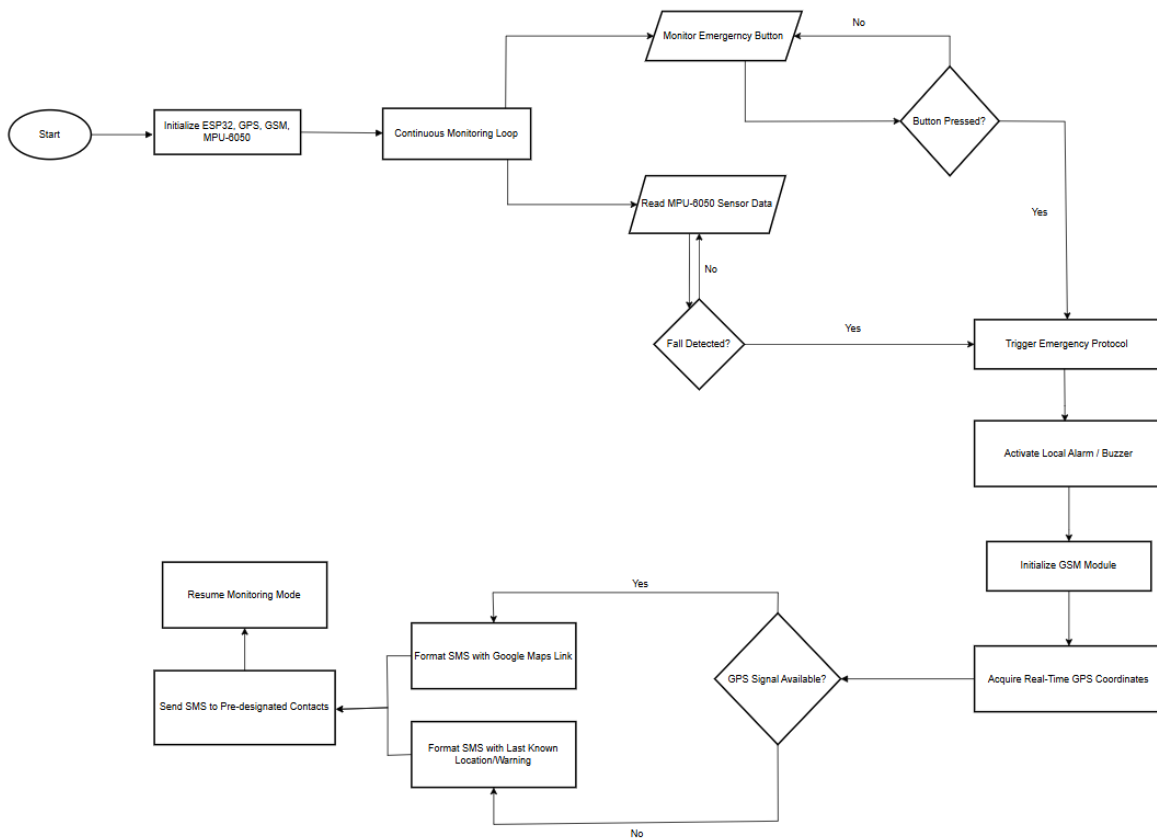


Figure 3: Flowchart

Flowchart Explanation:

- **Initialization:** The system booted up and initialized the core microcontroller (ESP32) and the peripheral modules: GPS, GSM and the MPU-6050 accelerator.
- **Monitoring:** Two inputs are constantly tracked by the device, including the physical condition of the emergency push button and the orientation information of the MPU-6050 sensor.
- **Decision Logic (Triggers):** Manual Trigger: in case of the user pressing the tactical button, the emergency protocol is activated. Auto Trigger: In case the accelerator records a sudden movement on the axis, which is indicative of a fall, the emergency procedure is triggered.
- **Response (Action):** Local Alarm: The piezo buzzer is triggered immediately to warn the people nearby and scare off attackers.
- **Location Tracking:** This module inquires the NEO-6M GPS to get the present latitude and the longitude.
- **Communication:** The SIM800L/SIM7000 GSM module transmits an SMS with the GPS coordinates (mostly in the form of a map link) to the emergency contacts of the user.

2.2.3. Circuit Diagram

A circuit diagram is a visual device showing the physical electrical connections of components of an IoT device or a system. (McRoberts, 2020) It depicts the electrical relationships between sensors, microcontrollers, actuators, power supplies and communication modules, and these may contain wires, pins, resistors, and other electronic parts. The actual IoT hardware is assembled, tested and troubleshooted with circuit diagrams, to ensure that the system is functional, and these are more detailed and technical than block diagrams or schematics.

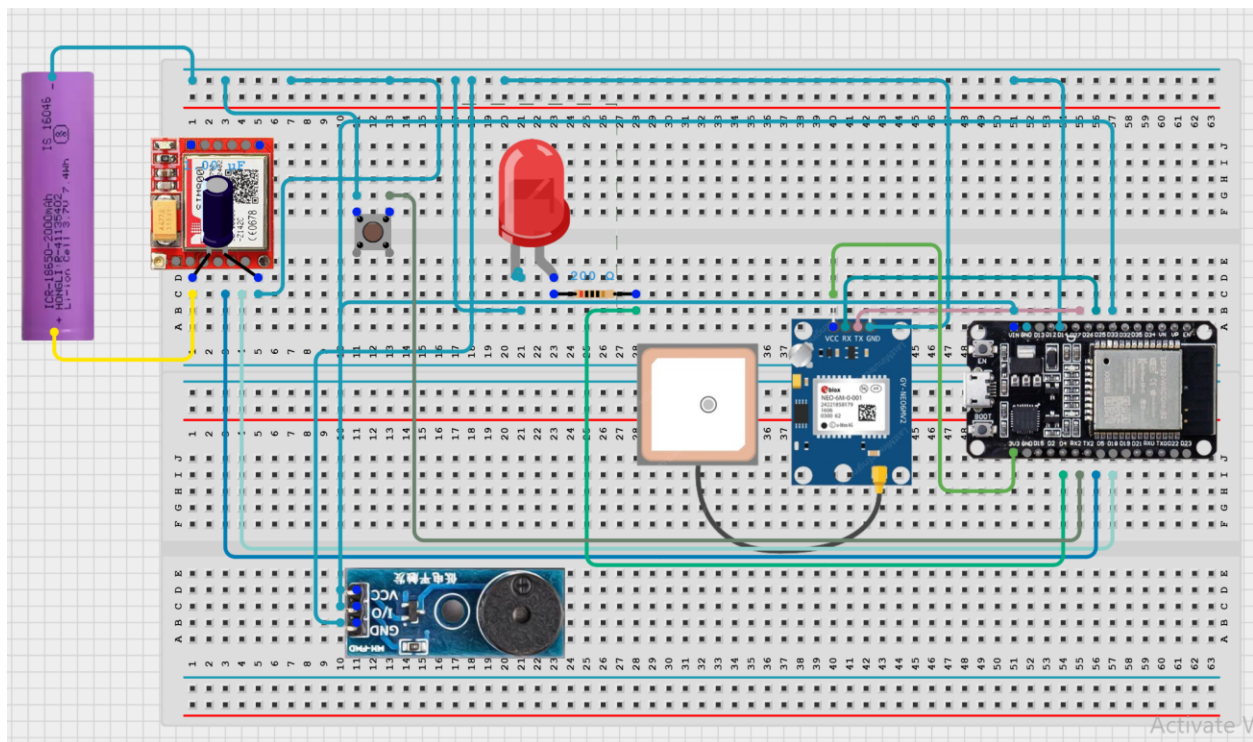


Figure 4: Circuit diagram

2.2.4. Block Diagram

An IoT block diagram is a graphical representation of an IoT system that presents the key functional units of such a system and their connection to each other. Characteristic blocks found in it include sensor blocks, data collection blocks, microcontroller processing, communications block, storage and analysis on cloud or server platforms, and user interfaces to monitor or control. (Greengard, 2021) A block diagram is meant to give a concise and simplified view of the system architecture in which it is easy to comprehend the working mechanism of an IoT solution without involving the circuit connections.

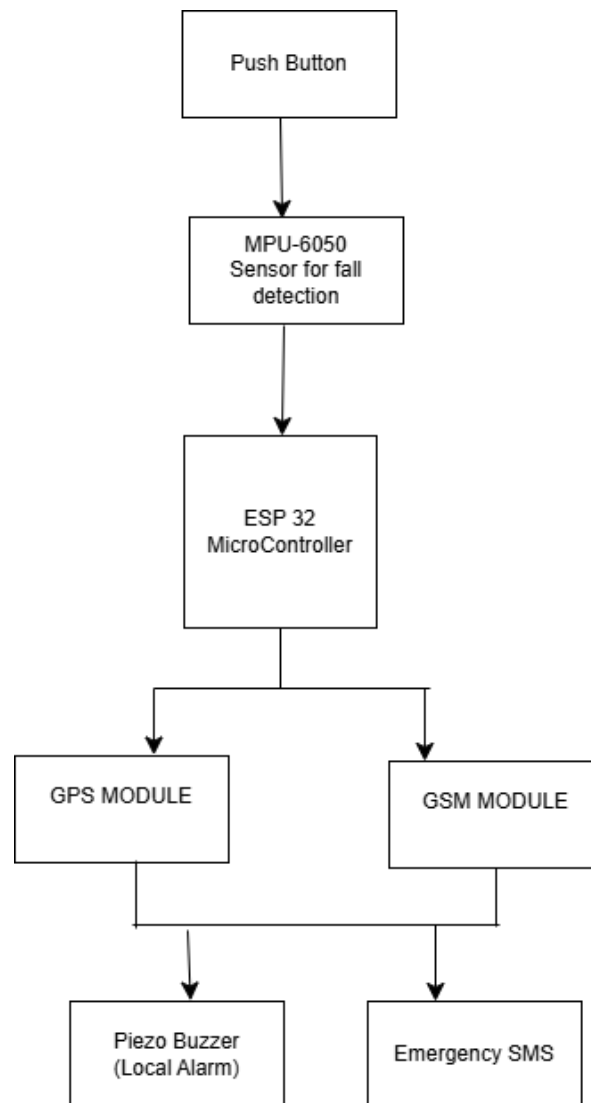


Figure 5: Block diagram

1. Input Layer

- **Push Button:** It is for conscious control. In case user is in danger they can press the button, and this triggers the alert immediately.
- **MPU-6050:** This is used for automatic detection. When the bracelet fall or thrown by user this sensor will get activated and inform the system that an accident has occurred.

2. Processing Layer

- **ESP 32 Microcontroller:** It is most significant component and is located in middle which listens input and coordinate with output. It ensures conversion of physical event into digital action.

3. Communication Layer

- **GPS Module:** It is for the coordination with ESP32 to get location. GPS gives the actual coordinates of latitude and longitude.
- **GSM Module:** It receives the location data that ESP32 is sending, and GSM module connects to mobile network through SIM Card.

Output Layer

- **Piezo Buzzer:** A loud sound is activated. This has two purposes; it scares away an attacker and lets anyone nearby know that someone needs help.
- **Emergency SMS:** A text message is received on the mobile of family. Since it has Google Map connection in GPS they can view the real time position of the user.

2.2.5. Schematics

Schematics define the elaborate drawing that displays the electrical linking and wiring of all elements in a circuit or apparatus. In contrast to a block diagram, which merely represents the functional components, a schematic contains the real components e.g. resistors, sensors, microcontrollers, LEDs and the precise connections which exist between them usually with representations of each component. (Greengard, 2021) It is also utilized in the design, construction, and troubleshooting of IoT hardware, so that all component parts are properly assembled to make the system physically functional.

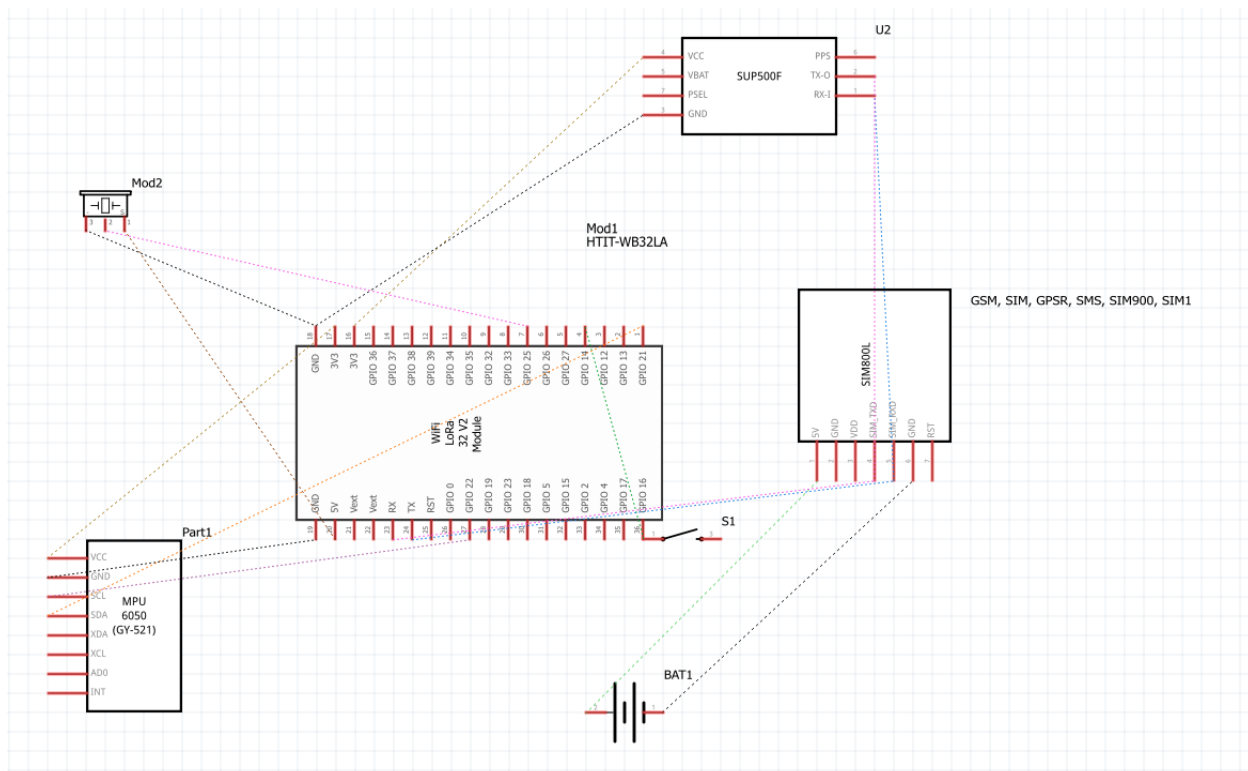


Figure 6: Schematic diagram

2.3. Requirement Analysis

Hardware resources:

- **Microcontroller: ESP32**

The primary microcontroller that manages the system is the ESP32. It links to external devices such as the GPS and the GSM modules, retrieves sensor data, and manipulates data.



Figure 7: ESP 32

- **GSM Module**

The GSM module is used to provide mobile connectivity with a SIM card. Whenever an alert is triggered, it lets the device send SMS messages to specified contacts.



Figure 8: GSM Module

- **GPS Module**

The GPS module provides real-time geographic co-ordinates to the device. It displays normal NMEA words together with the latitude and longitude following a constant extract of satellite messages.



Figure 9: GPS Module

- **MPU6050 Accelerometer**

The MPU6050 is employed to detect abnormal movement pattern, collision and unforeseen falls. It transmits accelerator and gyroscope data through the I2C interface and the ESP32 processes these data to determine the occurrence of a fall.



Figure 10: MPU6050 Sensor

- **Push Button**

The primary manual way of interaction with the user is the push button. On pressing, the device will trigger the emergency notification chain by transmitting a digital signal to ESP32.



Figure 11: Push button

- **Electrolytic capacitor**

The capacitor ensures that the power supply of the GSM module is stabilized. GSM units when transmitting SMSs can consume large burst currents making sudden voltage drop a possibility.

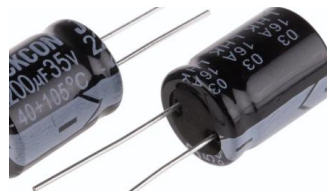


Figure 12: Electrolytic capacitor

Software resources:

- **Arduino IDE**

Arduino IDE is the main software used to write, compile and upload the code into microcontrollers such as ESP32 as in this case.

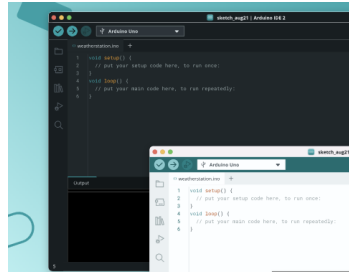


Figure 13: Arduino IDE

- **Figma**

Figma is a cloud-based design tool that was used to create block diagrams and system architecture.



Figure 14: Figma

- **Fritzing**

Circuit designer is a web-based application program helping simulate different circuit diagrams.

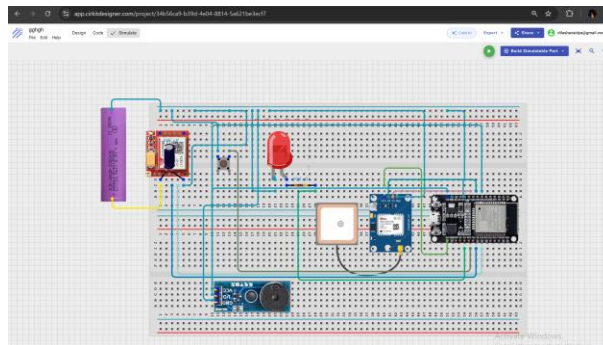


Figure 15: Circuit Designer

3. Development

3.1. Design and Planning

The first stage of development was dedicated to the development of the strong and practical design based on the virtual simulation and careful planning. The software, Cirkuit and Figma was initially used to model the system architecture and perform virtual assembly as well as validation of all the core hardware elements before prototyping. This was essential to avoiding wiring errors as well as to guarantee electrical compatibility.

To demonstrate this proof-of-concept, the design was made around five key hardware components, namely; the ESP32 microcontroller (as the central processing unit), a NEO-6M GPS module to obtain the location, a SIM800L GSM module to provide cellular access, an MPU-6050 accelerator/gyroscope to detect falls, and a tactile push button to activate the emergency manually. The local alarm outputs were a piezo buzzer and an LED. A breadboard was assigned as the prototyping platform to make life easier in terms of interconnection and testing.

The circuit connections were planned to use a rational flow of power and signals. This virtual design and connection planning stage gave an accurate blueprint and enhanced appreciation of the inter-module communication within the system. It laid a sure groundwork on the actual assembly of the body that would come later making the transition between simulation and a physical prototype to be effective and organized.

3.2. Resource Collection

Our project is an IoT-based women's safety device mechanism, which could not be developed without all the necessary gadgets. We accomplished this by completing research, then comparing prices, and then deciding after visiting a reputable and cost-effective wholesale.

Initially, we searched the internet for potential goods providers. We looked for things initially in Daraz, then approached other wholesalers in Ason Kathmandu. We went to the Ason market to check the availability of devices, affordable costs, and component requirements.

By purchasing the devices from a single supplier, our team was able to stay within budget while still ensuring that the components were equivalent. Our project now has access to all the resources that will be needed during the development and testing stages.

3.3. System Development

Phase 1: The SIM800L Module was tested. First, the lithium-ion battery was charged and connected to the SIM module via a capacitor to pass constant power to the module, The module was also connected to ESP and code was uploaded to the microcontroller.

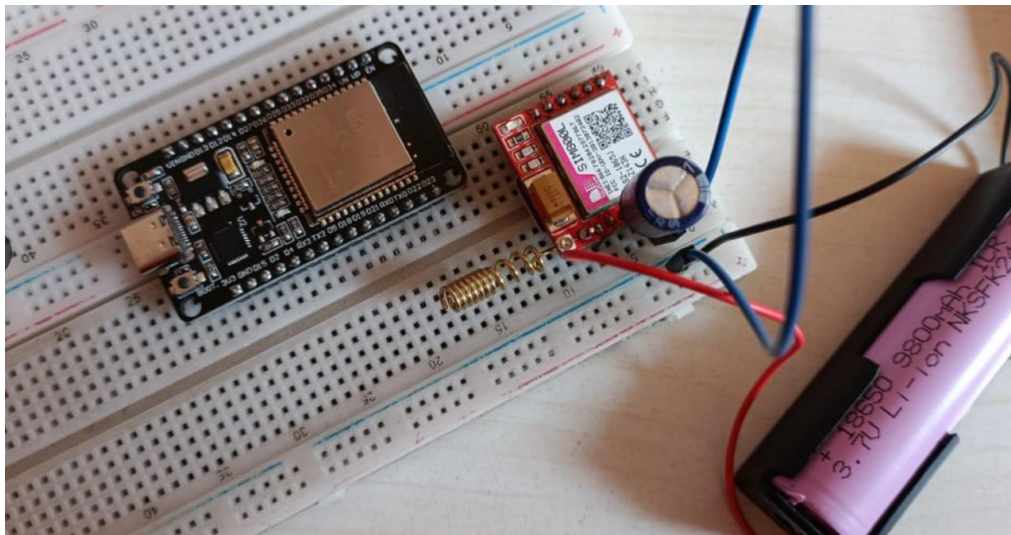
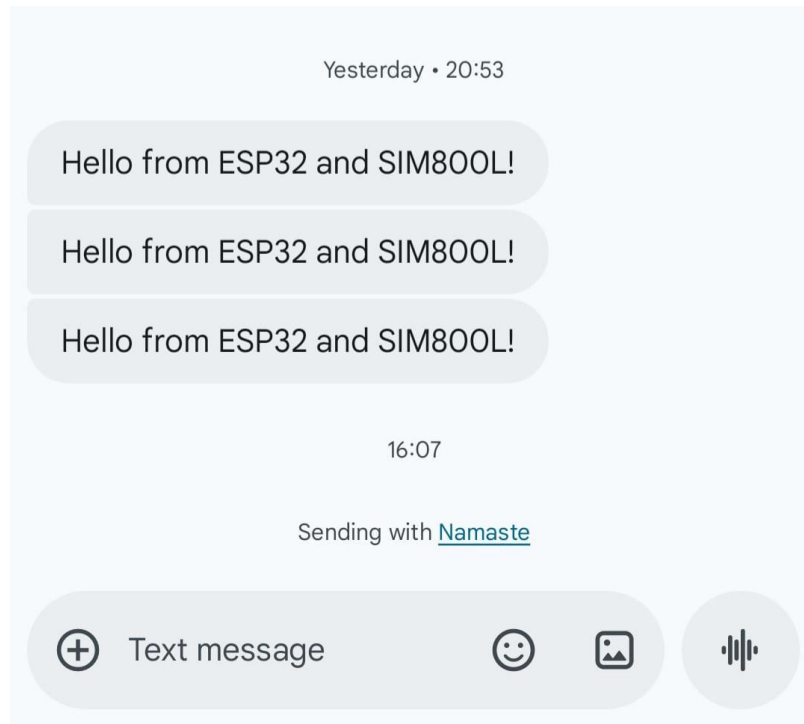
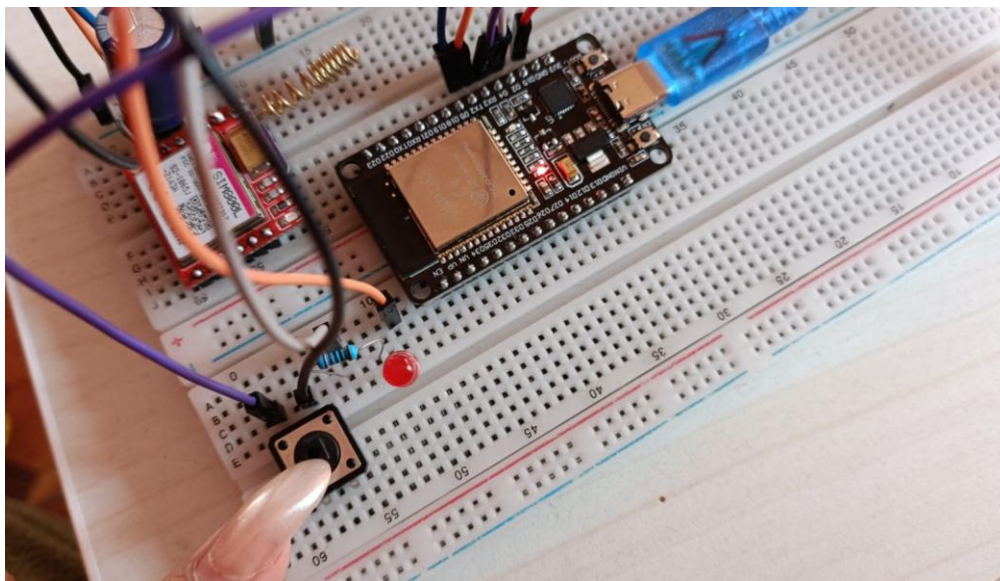


Figure 16: Testing the SIM Module



Phase 2: The Press button was tested using LED since, a picture wouldn't portray the buzzer sound. The necessary connections to ESP, LED and Press button were made. Then, the code was uploaded to ESP. On pressing the button, 0 as high signal was sent and the LED lit up.



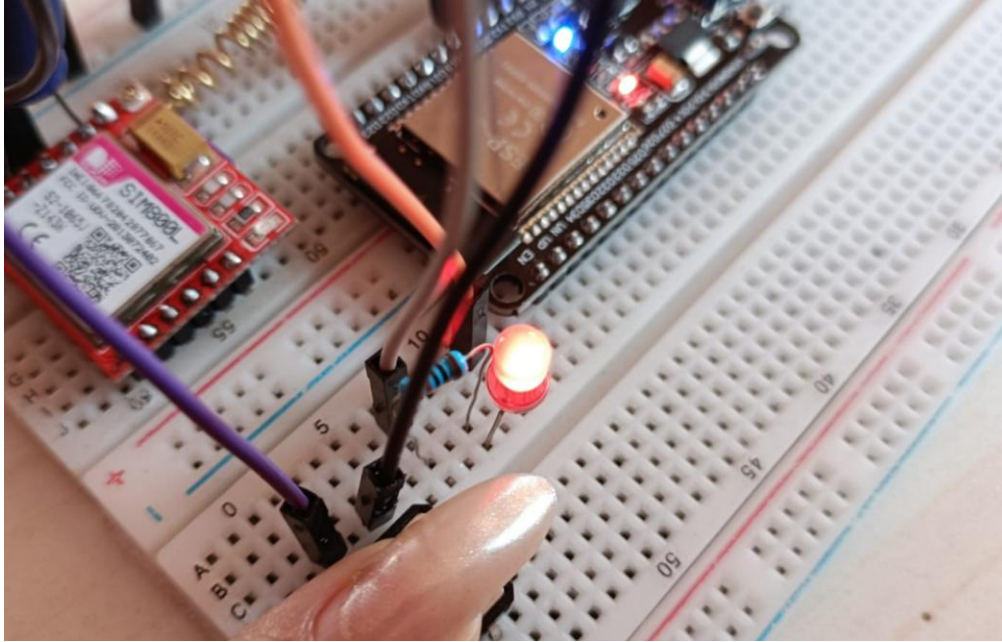


Figure 17: Testing the press button through LED

Phase 4: Adding the connections to ESP32, SIM800L Module and GPS Module on the breadboard.

Connections:

1. SIM800L Module

Table 1: Connection for SIM800L module

VCC	The positive terminal VCC was connected to the positive wire (red) of the lithium-ion battery
RXD	Receiver data point was connected to TX2 of ESP 32
TXD	Transmitter data point was connected to RX2 of ESP 32
GND	GND was connected to the negative (black) wire of the lithium-ion battery

2. GPS Module

Table 2: Connection pins for GPS Module

VCC	The positive terminal VCC was connected to 3.3V pin in ESP32
RXD	GPIO pin 27 of ESP 32
TXD	GPIO pin 26 of ESP 32
GND	GND was connected to the common ground in the breadboard

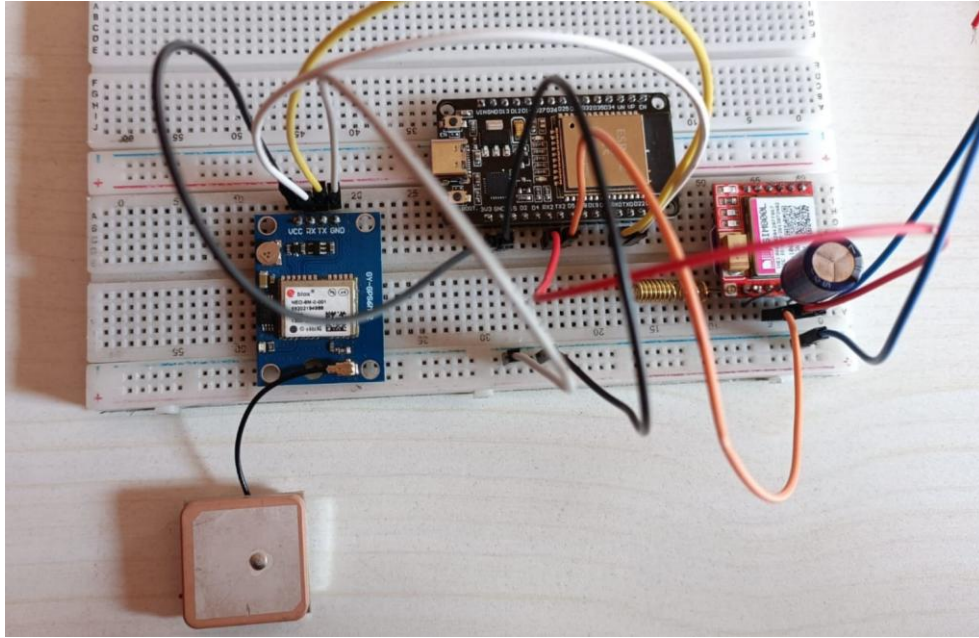


Figure 18: Connecting GPS, GSM and ESP on breadboard

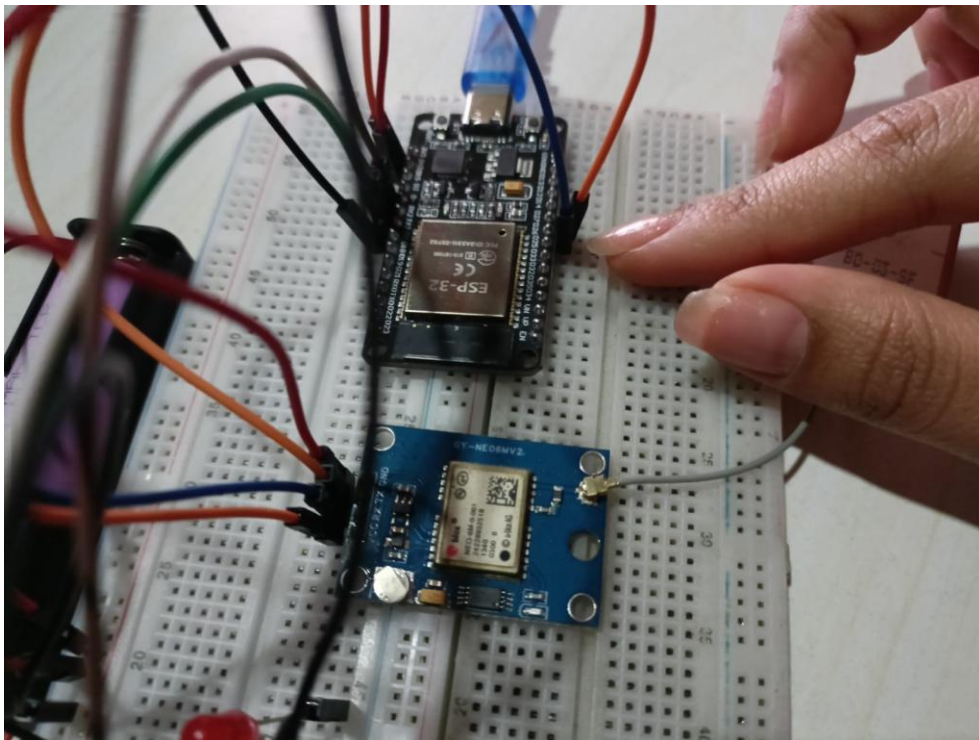


Figure 19: GPS module pin configuration

Phase 5: Adding press button and piezo buzzer to the breadboard.

Connection pins:

1. Press Button

Table 3: connection pins for Press button

VCC	The positive terminal VCC was connected to the D4 pin of ESP32
GND	GND was connected to the common ground line on the breadboard

2. Piezo Buzzer

Table 4: Connection pins for Piezo Buzzer

VCC	The positive terminal VCC was connected to the 5V Pin on ESP32
I/O	Receiver data point was connected to D25 pin on ESP32
GND	GND was connected to the common ground line on the breadboard

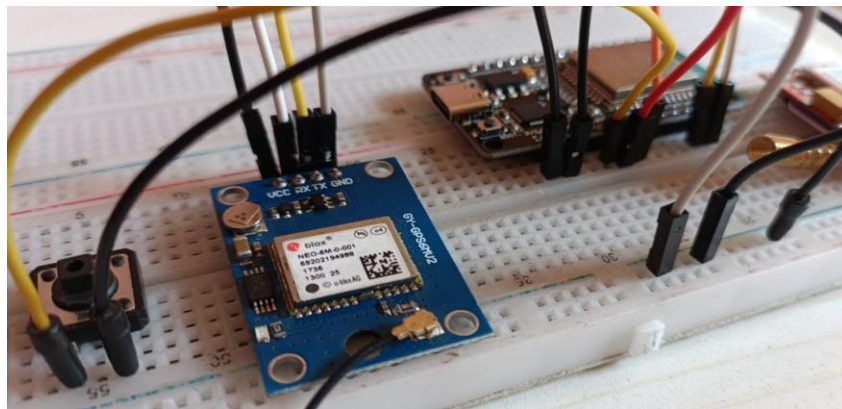


Figure 20: Adding the press button

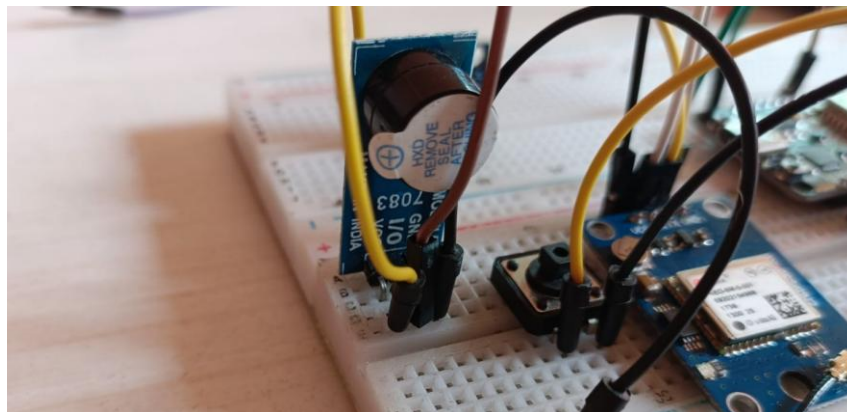


Figure 21: Adding the piezo buzzer

Phase 6: Necessary coding for the entire mechanism of SIM module, GPS module, LED and the piezo buzzer was done and saved on a single sketch in Arduino IDE.

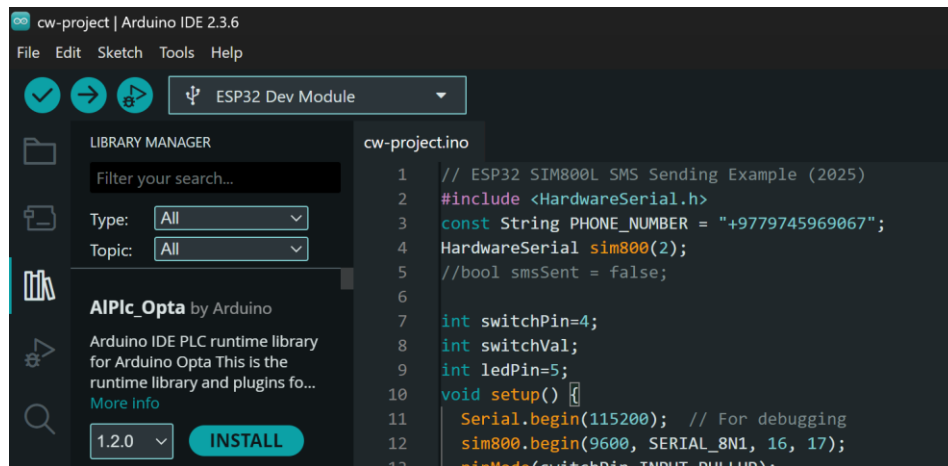


Figure 22: Coding on IDE

Phase 7: After all the connections of the modules were made successfully. The ESP microcontroller was connected to the laptop via a cable, and the code was compiled and uploaded. On successful upload, the button was pressed and that caused an SMS to go along with the location longitude, latitude and google map link. The LED also lit up.

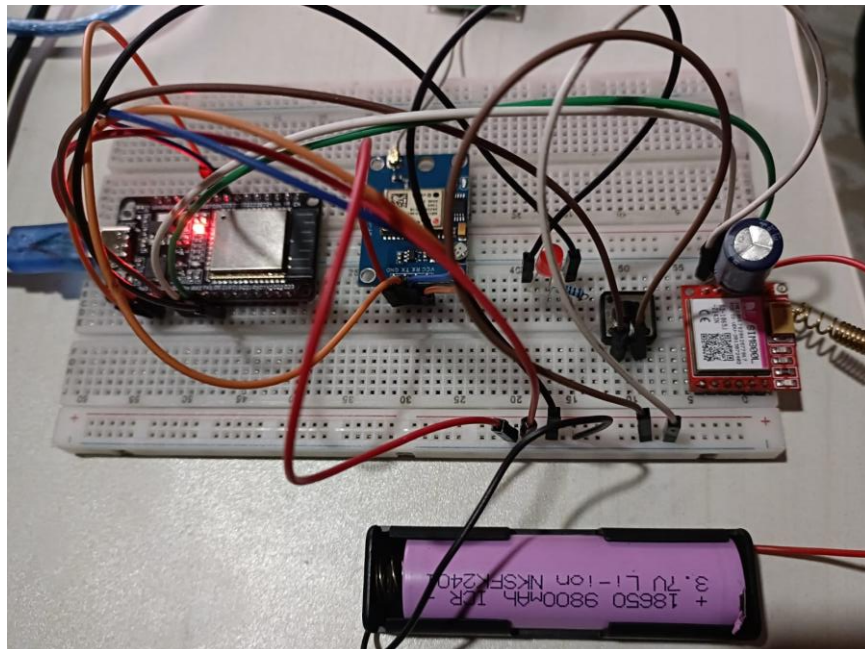


Figure 23: Overall circuit configuration

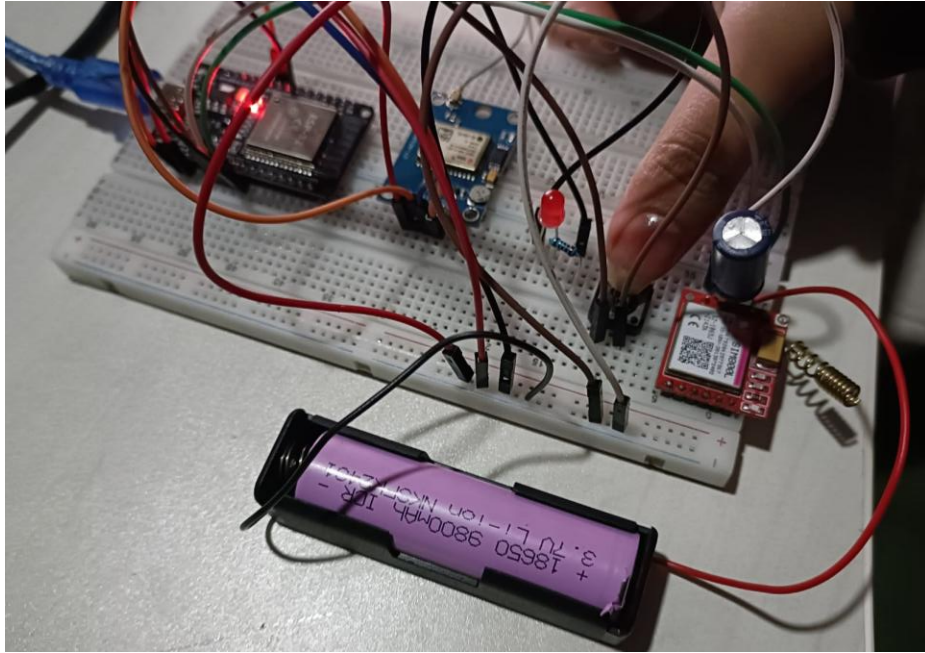


Figure 24: Pressing the trigger button

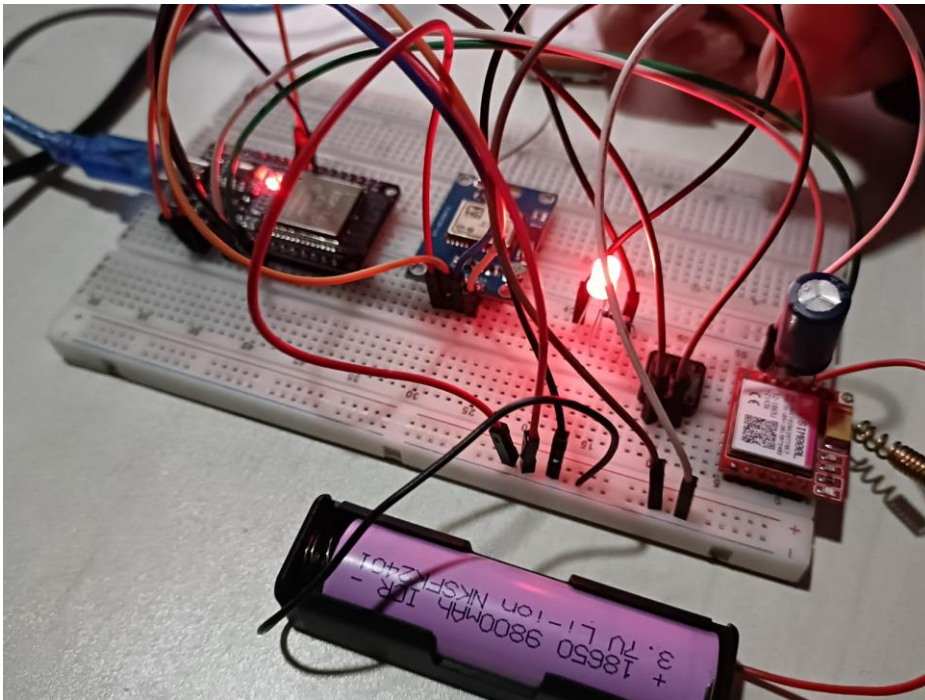


Figure 25: LED lights up on triggering

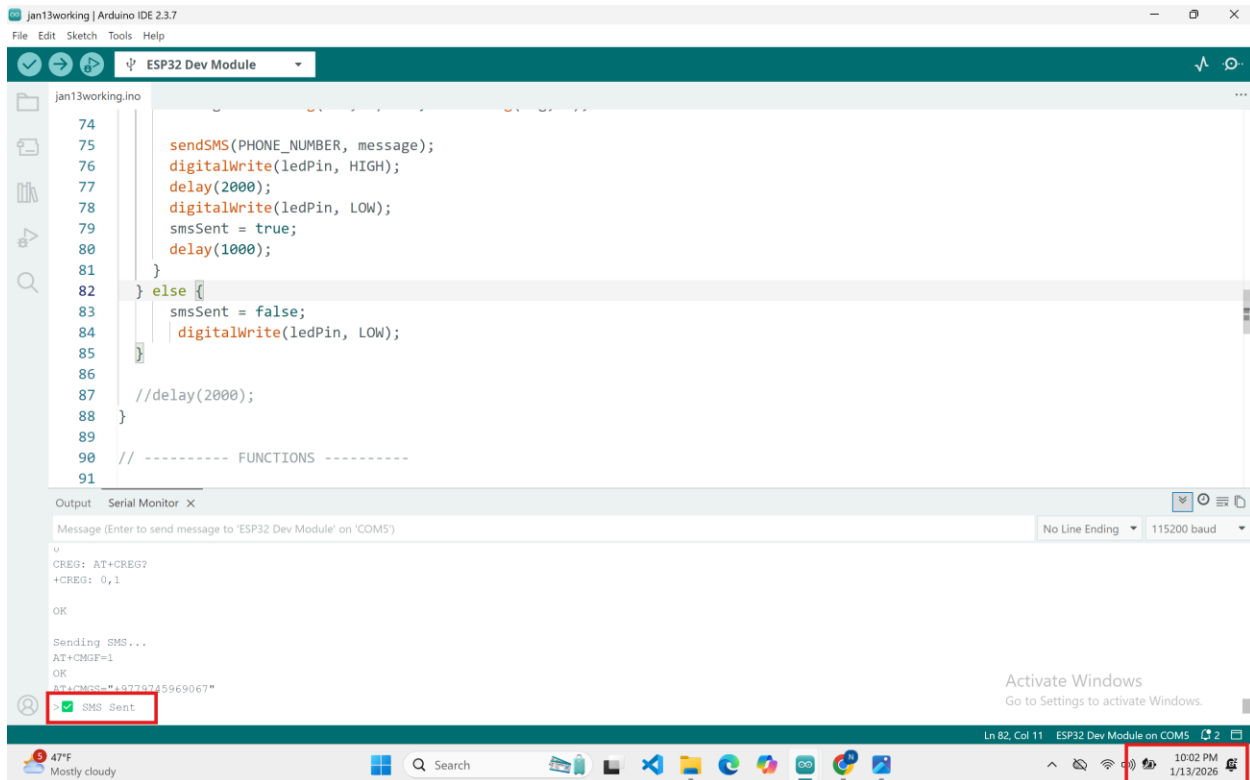


Figure 26: Serial monitor view

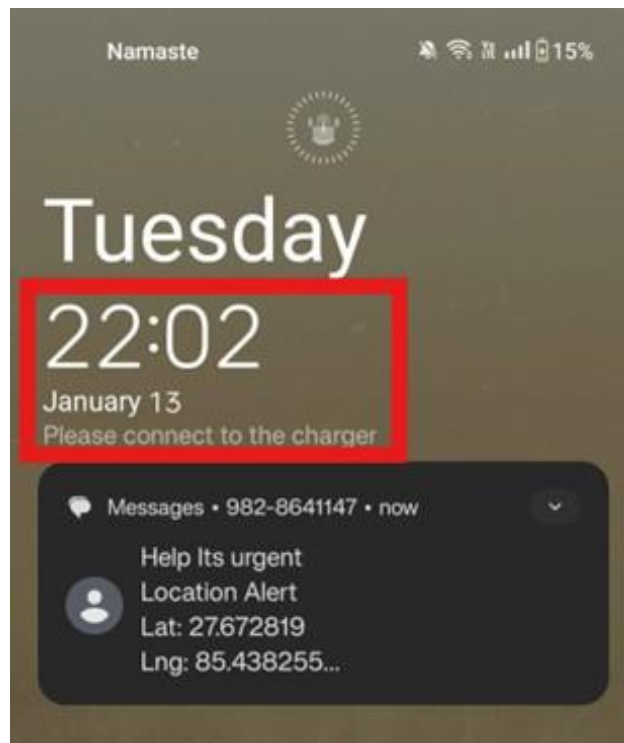


Figure 27: The SMS notification pops up

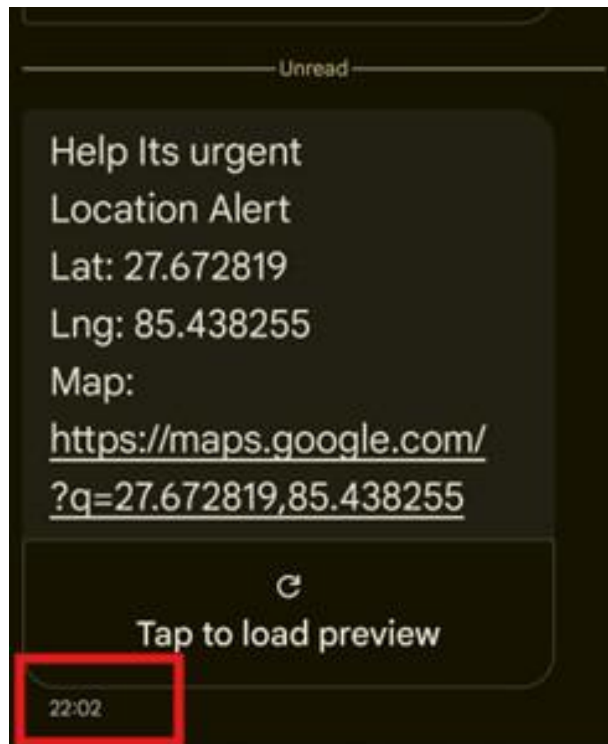


Figure 28: Full SMS view

4. Results and Findings

At the end of the development and testing processes, the prototype was able to prove its essence by serving as a reliable emergency alert system. The proposed IoT architecture was verified by the functionality of the integrated circuit and firmware. The system responded appropriately to manual and automated triggers.

The alert was triggered by a continued press of the emergency button or a false alarm of a simulated fall caused by the MPU-6050 sensor. This was used to activate the local audiovisual alarm (buzzer and LED) and the communication protocol. The GPS module continuously obtained the coordinates of the location which was packaged and sent as a live alert via the GSM module in form of an SMS with a live link to Google Maps. The process of activation through to confirmed receipt of the SMS by the test contacts was end-to-end, stable, and repeatable.

One technical observation was the necessity of controlling sequential starting of modules within the code to allow time to pass by the GPS to receive a location fix before the GSM tried to transmit. The dual-trigger mechanism (manual and sensor-based) proved to be a solid design feature, which brings redundancy to the various emergency situations. The project therefore provided a demonstrable piece of proof-of-concept that can be discreetly worn and enabled to call upon help by providing immediate local response as well as remote notification of pertinent location information.

Testing

Test 1:

Table 5: Test 1

Test No:	1
Objective	To test if the SIM800L Module works or not
Action	The required connections for the module were made and code to send message was uploaded to ESP32
Expected Result	An SMS appears on the added phone number
Actual Result	An SMS was sent successfully to the phone number
Conclusion	The test was successful

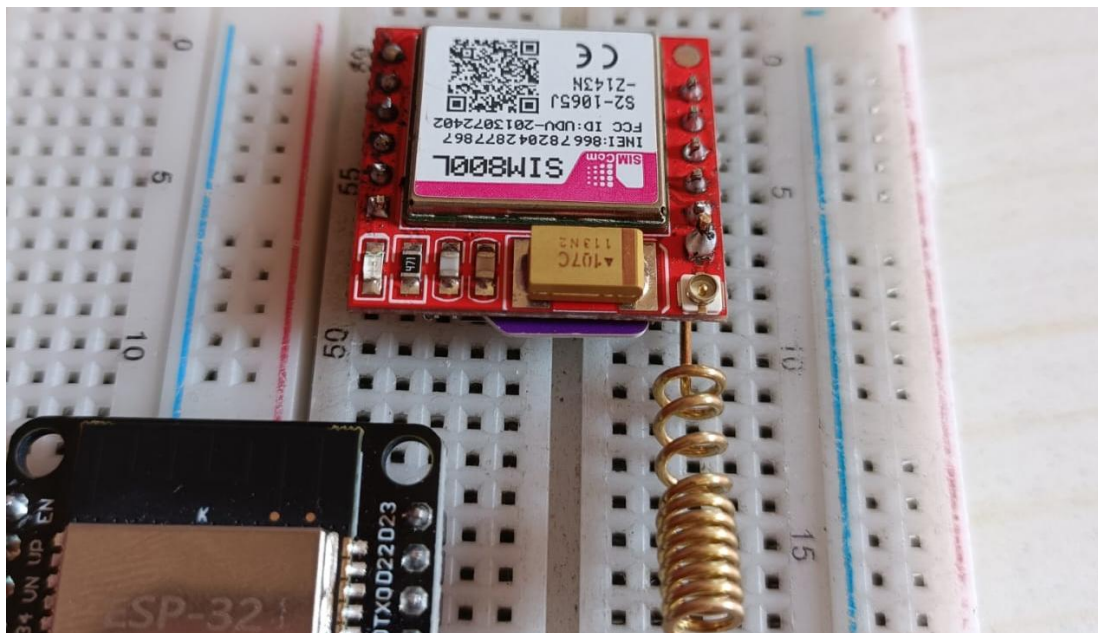


Figure 29: SIM800L Module

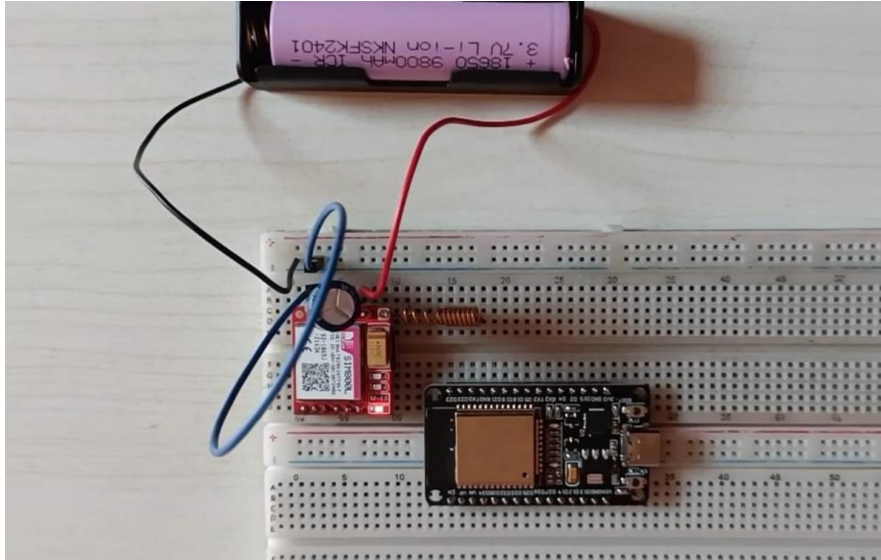


Figure 30: Configuring the GSM Module

The ESP32 was then connected to the laptop and the code to send a message was uploaded. Then the following message was obtained:

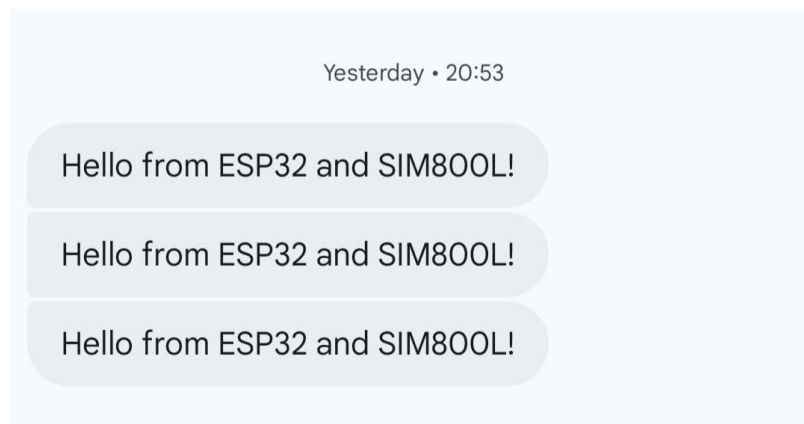
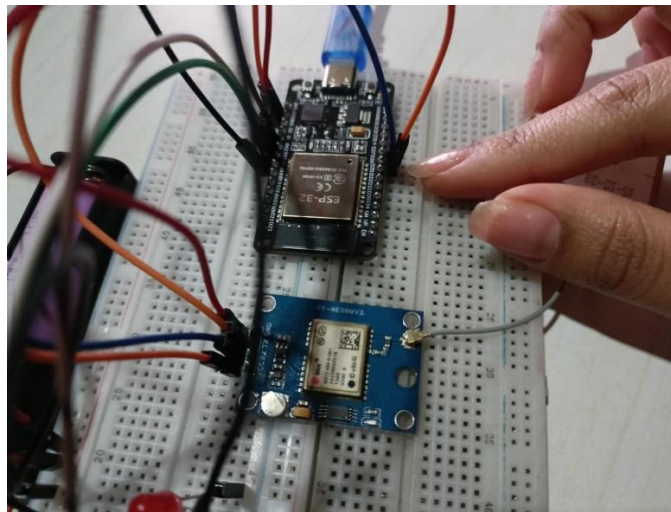


Figure 31: Expected result from the GSM Module

Test 2:*Table 6: Test 2*

Test No:	2
Objective	To test if the GPS Module works or not
Action	The GPS module was supplied suitable voltage and was kept under clear sky
Expected Result	The LED on the module blinks and on connecting location is retrieved
Actual Result	The GPS module was initiated, and longitude latitude was displayed.
Conclusion	The test was successful.

*Figure 32: Testing GPS module*

```

jan13working.ino
1 #include <HardwareSerial.h>
2 #include <TinyGPSPlus.h>
3
4 //OBJECTS
5 HardwareSerial sim800(2); // UART2 → SIM800L
6 HardwareSerial gpsSerial(1); // UART1 → GPS
7 TinyGPSPlus gps;
8
9 // SETTINGS
10 const String PHONE_NUMBER = "+9779745969067";
11
12 const int switchPin = 4;
13 const int ledPin = 5;
14 int switchVal;
15 bool smsSent = false;
16
17 // SETUP

```

Output Serial Monitor X

Message (Enter to send message to 'ESP32 Dev Module' on 'COM5')

No Line Ending 115200 baud

```

1
Lat: 27.672910
Long: 85.438089
1
Lat: 27.672918
Long: 85.438089
1
Lat: 27.672918
Long: 85.438089
1
Lat: 27.672910

```

Activate Windows
Go to Settings to activate Windows.

Figure 33: Expected result from the GPS module

5. Future Works

The Safety Bracelet prototype provides a working base and sets the direction towards development with well-defined ways of practice. The next task is to turn the proof-of-concept into a viable consumer product, which will happen in the future.

Key improvements include:

- A very suitable motion sensor could be configured into the circuit to detect the drastic motion or fall of the device which would then activate the trigger.

MPU 6050 Motion sensor

Table 7: Connection pins for MPU6050 sensor

VCC	The positive terminal VCC was connected to the 3.3V pin of ESP32
GND	GND was connected to the common ground line on the breadboard
SCL	The SCL pin was connected to the D22 pin on ESP32
SDA	The SDA pin was connected to the D21 pin on ESP32

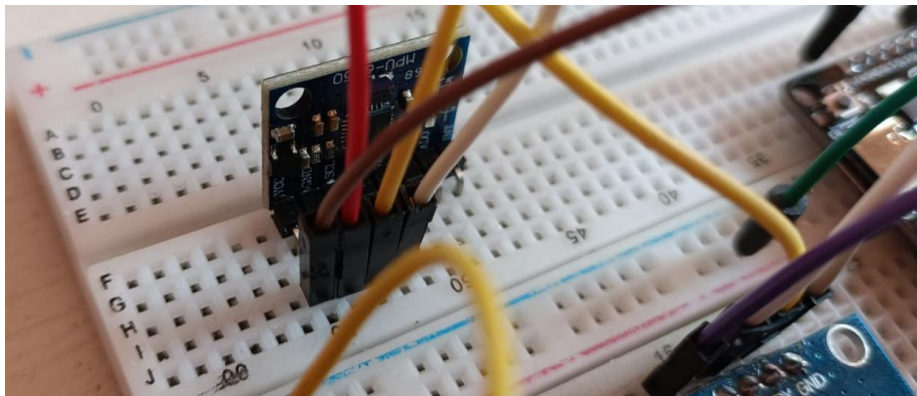


Figure 34: Adding connections to MPU 6050 sensor

- Hardware Miniaturization: Creating a custom PCB to fit all the components into a small and low-energy module to fit into a wearable form factor.
- Improved connectivity: The inclusion of the Bluetooth Low Energy (BLE) to pair with smartphones and configure the settings in the app.
- Intelligent Detection: The application of simple machine learning to sensor data to enhance the accuracy of fall-detection and minimize false alarms.
- Strong Ecosystem: Building a safe companion mobile application of real-time tracking and exploration integrations with emergency service API. The project is to be developed as a discreet, reliable, and smart wearable.

The device can be moved beyond the prototype to become a valid safety appliance by focusing on miniaturization, user-friendly design, and extremely high security to provide an immediate and reliable source of personal security.

6. Conclusion

The proposed project has been able to design, develop, and test a working prototype of an Internet of Things personal safety device, the Safety Bracelet. The essence of having a sound technical infrastructure with regard to instantaneous emergency notifying and location transmission has been fulfilled.

The integrated prototype showed that its dual-trigger system, which includes both manual activation and sensor-sensory fall detection, works and is implemented consistently in the entire alert sequence: the triggering of a local alarm, obtaining GPS coordinates, and sending an automated SMS to designated contacts. The results verify the feasibility of a purpose-specific hardware system that is not dependent on a smartphone and fills a serious void in the existing technology of personal safety.

The premeditated direction of virtual design and planning to the actual implementation and test has led to a functional model that forms a strong foundation. This project will eventually make an important and useful contribution to safety technology. It demonstrates that a small, wearable product can provide a more efficient, less noticeable, and dependable way of calling an ambulance, which is a good foundation towards future development into a commercially viable and life-saving item.

Appendix

```
#include <HardwareSerial.h>

#include <TinyGPSPlus.h>

HardwareSerial sim800(2);    // UART2 → SIM800L

HardwareSerial gpsSerial(1); // UART1 → GPS

TinyGPSPlus gps;

const String PHONE_NUMBER = "+9779745969067";

const int switchPin = 4;

const int ledPin = 5;

int switchVal;

bool smsSent = false;

// SETUP

void setup() {

    Serial.begin(115200);

    // SIM800L: RX=16 TX=17

    sim800.begin(9600, SERIAL_8N1, 16, 17);

    // GPS: RX=27 TX=26

    gpsSerial.begin(9600, SERIAL_8N1, 27, 26);

    pinMode(switchPin, INPUT_PULLUP);

    pinMode(ledPin, OUTPUT);
```

```
Serial.println("Initializing SIM800L...");  
delay(3000);  
  
sendAT("AT");  
sendAT("AT+CSQ"); // Signal quality  
sendAT("AT+CREG?"); // Network registration  
sendAT("AT+CMGF=1"); // SMS text mode  
  
Serial.println("Waiting for GPS fix...");  
}  
  
void loop() {  
  while (gpsSerial.available()) {  
    gps.encode(gpsSerial.read());  
  }  
  if (gps.location.isValid()) {  
    Serial.print("Lat: ");  
    Serial.println(gps.location.lat(), 6);  
    Serial.print("Lng: ");  
    Serial.println(gps.location.lng(), 6);  
  } else {  
    Serial.println("No GPS fix yet...");  
  }  
  switchVal = digitalRead(switchPin);
```

```
Serial.println(switchVal);  
if (switchVal == 0 && !smsSent) {  
  
    if (gps.location.isValid() && isNetworkRegistered()) {  
  
        float lat = gps.location.lat();  
        float lng = gps.location.lng();  
  
        String message = "Help Its urgent\n";  
        message += "Location Alert\n";  
        message += "Lat: " + String(lat, 6) + "\n";  
        message += "Lng: " + String(lng, 6) + "\n";  
        message += "Map: https://maps.google.com/?q=";  
        message += String(lat, 6) + "," + String(lng, 6);  
  
        sendSMS(PHONE_NUMBER, message);  
        digitalWrite(ledPin, HIGH);  
        delay(2000);  
        digitalWrite(ledPin, LOW);  
        smsSent = true;  
        delay(1000);  
    }  
} else {  
    smsSent = false;  
    digitalWrite(ledPin, LOW);
```

```
}

//delay(2000);

}

// Send SMS

void sendSMS(String number, String message) {

    Serial.println("Sending SMS...");

    sim800.println("AT+CMGF=1");

    delay(500);

    sim800.print("AT+CMGS=\"");
    sim800.print(number);
    sim800.println("\");

    if (!waitForPrompt('>', 10000)) {
        Serial.println("Failed to get '>' prompt");
        return;
    }

    sim800.print(message);
    sim800.write(26);

    Serial.println("SMS Sent");
}
```

```
// Check network registration

bool isNetworkRegistered() {
    sim800.println("AT+CREG?");
    delay(500);

    String response = "";
    while (sim800.available()) {
        response += char(sim800.read());
    }

    Serial.println("CREG: " + response);

    return (response.indexOf(",1") != -1 || response.indexOf(",5") != -1);
}

bool waitForPrompt(char prompt, unsigned long timeout) {
    unsigned long start = millis();
    while (millis() - start < timeout) {
        while (sim800.available()) {
            char c = sim800.read();

            Serial.write(c);

            if (c == prompt) return true;
        }
    }

    return false;
}
```

```
}  
void sendAT(const char *cmd) {  
    Serial.print(">> ");  
    Serial.println(cmd);  
    sim800.println(cmd);  
    delay(800);  
    while (sim800.available()) {  
        Serial.write(sim800.read());  
    }  
    Serial.println();  
}
```

References

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McRoberts, M., 2020. *Beginning Arduino*. 2nd edition ed. New York: Apress.

Sharma, A. & S. P., 2022. *IoT Based Women Safety Device with real-time GPS Tracking*, s.l.: International Journal Of Engineering Research.

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